C64uRemote for M5Stack Core

User Manual

Version 1.0

Contents

Welcome

C64uRemote turns an **M5Stack Core** into a convenient remote control for the **Commodore 64 Ultimate (c64u)** and the **Ultimate64 Elite-II**. Over Wi-Fi you control the machine remotely: reset, reboot, power off, open the Ultimate menu and change the CPU speed.

With the optional **RFID2 reader** and a **microSD card** it becomes a tap-to-launch console: you hold an NFC card to the device and the matching game starts on the C64. The cards are compatible with the TeensyROM and Zaparoo format – the same card works on both systems.

This project builds on the original idea by Karl Prosser (@klumsy) and was extended for the M5Stack Core by Martin Oswald (@mad, 1MHz.de).

What you need

Required:

- An M5Stack Core (Basic, Gray or Fire)
- A Commodore 64 Ultimate or Ultimate64 Elite-II on the same Wi-Fi

Optional, for the NFC features:

- M5Stack Unit RFID2 (WS1850S), connected to Port A
- A microSD card (formatted FAT32) in the Core
- NFC cards: NTAG215 recommended, NTAG213/216 and MIFARE Classic also work

Without the reader or SD card, all other functions keep working normally.

First-time setup

For the Core to find your C64, the Wi-Fi credentials and the C64's address must be stored once. You can do that **right on the device** under *SETUP → WiFi* – the source code no longer has to be touched. The credentials go into internal memory and survive every restart. The *Setting up Wi-Fi* chapter walks through the details.

If you would rather supply the credentials while programming the device, put them into `build_env.h` as before (see the technical documentation). They then serve as the initial values for the very first start.

The connection status is always shown in the top left of the display:

Display	Meaning
C64U OK (green dot)	Everything connected, ready
NO C64U (blue)	Wi-Fi present, but the C64 does not respond
AUTH? (yellow)	C64 reachable, but the password is wrong
NO WIFI (red)	No Wi-Fi connection

To the right you see the IP address, the current CPU speed and whether the RFID reader and SD card were detected.

The three buttons

The Core has three buttons below the display: **A**, **B** and **C**. The bottom line of the screen always shows what they currently do – for A and C as an arrow with the button letter, for B as text.

Button	Short press	Long press (from 0.6 s)
A	move selection left / up	back, or to the home screen
B	select / confirm	special function (configurable)

Button	Short press	Long press (from 0.6 s)
C	move selection right / down	special function (configurable)

The home screen

At the top the status bar, in the middle the animated logo, below it all commands as tiles:

```
RESET   REBOOT   MENU    POWER   CPU
RFID    SD       STATUS  SETUP
```

Use **A** and **C** to select a tile (it gets a bright outline), and **B** to trigger it.

Tile	What happens
RESET	The C64 is reset (like the reset button)
REBOOT	The C64 restarts completely
MENU	Opens or closes the Ultimate menu on the C64
POWER	Powers off the C64 – press twice for safety
CPU	View and change the CPU speed
RFID	Tap a card and start the game stored on it
SD	Pick a game directly from the SD card and start it
STATUS	Detailed connection info, connection test with B
SETUP	Settings and NFC tools

Powering off (POWER): Turning off by accident would be annoying, so the prompt *POWER OFF? NOCHMAL!* ("again!") appears first. Only a second press of **B** within the time window actually powers off.

Changing the CPU speed

Select the **CPU** tile and open it with **B**. The current speed is shown at the top, the list of options below. Use **A/C** to pick the desired speed, and **B** to set it. The Core reads the available steps directly from the C64, so you get exactly the values your device supports.

Launching games by card (RFID)

Requirements: RFID2 reader connected, a microSD with your games inserted, and the card written beforehand (see below).

Just present the card (automatic)

Normally you do not have to operate anything at all: while the main screen is showing, the Core checks in the background whether a card is present. As soon as it finds one it switches to read mode on its own and launches the stored program.

The reading screen then stays up for about 20 seconds, so you can present the next card right away. After that — or as soon as you press a button — it returns to the main screen.

How often it looks is set under *SETUP → Auto-NFC*:

Setting	Meaning
Off	no background polling, cards only through the RFID tile
1.5s	very frugal
0.7s	factory default, a good compromise
0.3s	quickest to react

The probe is short enough that even at *0.3s* it slows neither the animation nor the controls noticeably.

Through the RFID tile

If you want to read a card deliberately — or if *Auto-NFC* is switched off — the manual route still works:

1. Select the **RFID** tile and open it with **B**.
2. Place the NFC card on the reader.
3. The Core reads the stored path, fetches the file from the SD card and sends it to the C64. A progress bar shows the upload.
4. After *GESTARTET: ...* ("started") the game runs.

Remove the card, tap the next one – the screen stays in read mode, so you can play several cards one after another.

Which files work

Extension	What it is
`.prg`	Program (loaded and started)
`.crt`	Cartridge
`.sid`	SID music
`.mod`	Amiga MOD music
`.d64 .d71 .d81 .g64 .g71`	Disk images

For disk images the image is mounted as drive 8. What happens next is set under *Disk Action* in the setup: mount only, mount and reset, or mount and automatically start the first program.

Command cards

A card does not have to point at a game – it can also carry a **command**. Present it and the command runs immediately, with no menu and no SD card needed.

Card	Effect
Reset	Resets the C64
Reboot	Restarts the C64 completely
Ultimate Menu	Opens or closes the Ultimate menu
PowerOff direct	Powers off immediately
PowerOff with prompt	Asks first – present the same card a second time within the time window to confirm
CPU x MHz	Sets the CPU to the value stored on the card

Creating a command card

1. Choose **NFC-Cmd** in the settings.
2. Pick the command from the list. After the fixed entries come all CPU steps your C64 offers, so a "CPU 10 MHz" card is a single click.
3. Present the card; *KARTE OK* means written and verified.

PowerOff with prompt

The waiting time lives **on the card**, not in the device. When writing, the value comes from *NFC-Cmd PowOff* (3, 5, 8 or 15 seconds, 8 s by default). Presenting such a card shows *POWER OFF? NOCHMAL!* with a countdown. To power off:

- present the **same card** again, or

- press the **button**

If the countdown expires or a different card appears, nothing happens. A card with a time of **0** powers off immediately without asking.

What is stored on the card

The command is plain text in an NDEF record – you can inspect or write it with any NFC app on your phone:

```
CMD:RESET
CMD:REBOOT
CMD:MENU
CMD:POWEROFF=0      power off immediately
CMD:POWEROFF=8      ask first, 8 seconds to confirm
CMD:CPU=10          set the CPU to 10 MHz
```

Case does not matter. The M5Dial and M5Stack Core editions understand the same format, so one card works on both.

Launching games directly from SD (no card)

Open the **SD** tile. You see the files and folders on your SD card. This lets you start something without an NFC card – handy for trying things out.

In the SD menu the buttons are mapped specially, because you often scroll a lot here:

Button	Function
A / C short	one entry up / down
A long	one folder back
C hold	keeps scrolling automatically
B	start file (or open folder)
B long	back to the home screen

The title bar shows these special functions as a reminder.

Writing and managing NFC cards

All card tools are found under **SETUP** at the very top. Open the setup with the SETUP tile; the first four entries are the NFC actions.

NFC-Write: assign a game to a card

1. Open the setup; **NFC-Write** is already selected → **B**.
2. Select the desired game file on the SD card → **B**.
3. Place the NFC card on the reader.
4. The path is written and immediately read back for verification. *KARTE OK* means success.

The format is compatible with the TeensyROM/Zaparoo system. A card written this way works on both devices, as long as the folder structure on both SD cards is the same. You can also write cards with your phone (the *NFC Tools* app, text record).

NFC-Info: read out a card

Shows everything on the card: UID, card type, memory size, the stored path and whether the file actually exists on the SD card. Use **A** or **C** to page to **page 2**, which shows technical raw data, lock bits, access protection and the read counter.

Handy: this is how you find cards whose game you renamed or moved – the *Datei* (“file”) line then reports “NICHT auf SD” (“not on SD”).

NFC-Dump and NFC-Restore: copying cards

- **NFC-Dump** saves the complete card content as a text file in the folder `/NFC-DUMPS/` on the SD card. Works with any card, including foreign MIFARE cards without text.
- **NFC-Restore** writes such a dump back onto a (blank) card.

This lets you clone cards. One note: a card's serial number (UID) is fixed in the chip and cannot be copied – but the content is copied completely. For game cards that is enough, because what matters there is the stored path.

Setting up Wi-Fi

Everything for this sits under **SETUP → WiFi**. Pick an entry with **A** and **C**, trigger it with **B**. The Core remembers up to **four networks** and tries them one after another when connecting – handy if you move between home and a phone hotspot.

Entry	Effect
Scan networks	Search the area and pick a network from the list
Load from SD	Read `wifi.txt` from the microSD
Setup portal	Access point of its own with a web interface
Saved	Pick a stored network and connect
To NFC card	Write a stored network to an NFC card
Save to SD	Write all stored networks to the microSD as `wifi.txt`
Delete network	Remove a single entry
Delete all	Discard all credentials

Route 1: scan, then hand over the password on a card

Scan networks shows the networks found after a few seconds, along with their signal strength. A network that is already stored is marked *known* and connects straight away; an *open* network needs no password.

For everything else the Core asks for the password and waits for an NFC card. The card carries an ordinary text record in the same scheme that Wi-Fi QR codes use:

```
WIFI:S:MyWiFi;T:WPA;P:MyPassword;;
```

Write the card with a phone, for example with the **NFC Tools** app (*Write → Add a record → Text*). If the card holds only the password without the `WIFI:` prefix, it is assigned to the network you picked beforehand.

> The password sits unencrypted on the card. Once you are set up, overwrite the card – the device keeps the password in internal memory anyway.

Presenting a card is enough

A complete `WIFI:` card also works **outside** the setup: present it on the home screen and the Core stores the network and connects to it right away. It waits up to eight seconds and then reports *WIFI ACTIVE* with the IP address, or *NETWORK NOT THERE* if the network is out of range. If that network is already connected, nothing happens (*ALREADY CONNECTED*) – so the card may simply stay on the reader.

That gives a second device its credentials in a matter of seconds.

Route 2: a file on the SD card

Put a text file `wifi.txt` in the root directory of the microSD:

```
ssid = MyWiFi
pass = MyWiFiPassword

ssid = Hotspot
```

```
pass = secret123

host     = 192.168.0.64
hostpass =
```

Every new `ssid` line begins a new entry; lines starting with `#` are comments. The file is read at start-up (as long as no network is stored yet) and at any time through *Load from SD*.

The other way round, **Save to SD** writes all stored networks together with `host` and `hostpass` back out in exactly this format. An existing `wifi.txt` is renamed to `wifi.bak` first, so nothing is lost. That makes setting up a second device a no-typing affair – bear in mind that the passwords sit on the card in plain text.

Route 3: the setup portal

Setup portal turns the Core into an access point of its own for five minutes. The display shows the network name, the password and the address you open in a browser. There you enter SSID and password – either from the list of networks found or by hand – and optionally the address and password of the C64.

After saving, the Core shuts the access point down and connects to the new network. The browser connection breaking off in the process is normal.

Settings overview

In **SETUP**, after the NFC actions, you find all settings. They are stored permanently and reloaded after power-cycling.

NFC and Wi-Fi

Setting	Options
NFC-Random	Create a random-pick card for a directory
NFC-Cmd	Create a command card (see the *Command cards* chapter)
NFC-Cmd PowOff	Default prompt time for a PowerOff command card: 3, 5, 8, 15 s
WiFi	Wi-Fi setup submenu (see the *Setting up Wi-Fi* chapter)
Auto-NFC	Background polling interval: *Off*, *1.5s*, *0.7s*, *0.3s*

Button mapping

Setting	What it does
Taste B lang	Action for a long press on B
Taste C lang	Action for a long press on C
B lang + C	Action for the sequence B long, then C
B lang + A	Action for the sequence B long, then A

Each of these four can be set to **Off, Reset, Reboot, Menu** or **PowerOff**. Factory default: B long = Reset, C long = Menu, B+C = Reboot, B+A = off.

How the sequences work (one-handed): hold B until *LOSLASSEN, DANN A / C* (“release, then A / C”) appears, then release and briefly tap A or C. If you do nothing, the “B long” action runs after a short delay.

Setting	Options
PowerOff Kombi	Whether PowerOff via a key sequence prompts first (confirm with A) or powers off directly
PowerOff Zeit	How long the power-off prompt stays valid (0.5 to 3.0 s)
Kombi Zeit	How long the device waits for the second key after “B long” (0.5 to 3.0 s)

Display and effects

Setting	Options
Animations	Logo animations on/off
Effect	Which effect: Auto (cycle), Static, Water, RotoZoom, SineWave, Ripple, Raster
FX Detail	Half or Full – Half computes more coarsely but runs smoother (recommended on the Core Basic)
Anim Speed	Animation speed: Slow / Normal / Fast
Effect Time	How long an effect runs: Short / Normal / Long
Static Time	How long the calm logo is shown in between
Brightness	Display brightness (32 to 255)

C64 options

Setting	Options
Disk Action	After mounting a disk image: mount only, mount + reset, or mount + start the first program
Disk Drive	Target drive: Auto (bus 8), or fixed A / B

Miscellaneous

Setting	Options
Beep	Key click on/off
Factory Reset	Reset all settings to factory defaults

Frequently asked questions

The Core shows “NO WIFI”. Check that the Wi-Fi name and password are stored correctly and the router is in range. The Core retries the connection every few seconds.

“AUTH?” in the status bar. A network password is set on the C64 that is not stored on the Core. Check the password in the C64 menu or enter it in `build_env.h`.

A card starts nothing / “TYP UNBEKANNT” (unknown type). The Ultimate can only start the file types listed above. Use **NFC-Info** to check which path and extension are on the card and whether the file is on the SD.

The SD card is not detected. Format as FAT32, 32 GB maximum, and if needed try a different card. The Core automatically tries two speeds.

The PowerOff command reports an error. That is normal – when powering off the C64 often no longer responds cleanly.

Can I use the same cards on the TeensyROM? Yes. The card format is identical. The only important thing is that the folders on both SD cards have the same names.

License

How this was made

The port, the extensions and these manuals were written with the help of Claude (Anthropic). Concept, idea, hardware decisions and every test on real devices: Martin Oswald (@mad).